

3m



8.5

Chapter 3 + 6 Test Thinking

Name

For each question, show all of your work, draw FBD when

Rubric	Level 4+ 10 marks	Level 4 8-9 marks	Level 3 7 marks	Level 2 6 marks	Level 1 5 mark	NOT YET
Thinking/ Inquiry Questions answered using the formulas and units required to demonstrate understanding of the concepts.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of All concepts/ questions demonstrated.	All questions answered using the correct formulas and units as required. A thorough and correct understanding of Most concepts/ questions demonstrated.	All questions answered using most of the correct formulas. A good understanding of most concepts/ questions demonstrated.	Most questions answered or attempted. Questions answered using some of the correct formulas. Some understanding of concepts/ questions demonstrated.	Some questions answered or attempted. Questions answered using limited formulas. Limited understanding of concepts/ questions demonstrated.	Questions not answered or show little understanding of concepts.

1. Three asteroids are arranged in a right angle as shown below. Asteroid C has a mass of 3.0×10^{20} kg. Asteroid B has a mass of 1.2×10^{21} kg and is 6.5×10^{10} km away from asteroid C. Asteroid A has a mass of 5.0×10^{20} kg, and is 9.0×10^{10} km away from asteroid C. What is the net force of gravity pulling on object asteroid C?

Assume listed distances are centre of mass to centre of mass

Show all work, keep units throughout!

$$m_C = 3.0 \times 10^{20} \text{ kg}$$

$$m_B = 1.2 \times 10^{21} \text{ kg}$$

$$m_A = 5.0 \times 10^{20} \text{ kg}$$

$$r_{AC} = 9.0 \times 10^{10} \text{ km}$$

$$\rightarrow 9 \times 10^{13} \text{ m}$$

$$r_{BC} = 6.5 \times 10^{10} \text{ km}$$

$$\rightarrow 6.5 \times 10^{13} \text{ m}$$

$$\textcircled{1} F_{gAC} = \frac{G m_A m_C}{r_{AC}^2}$$

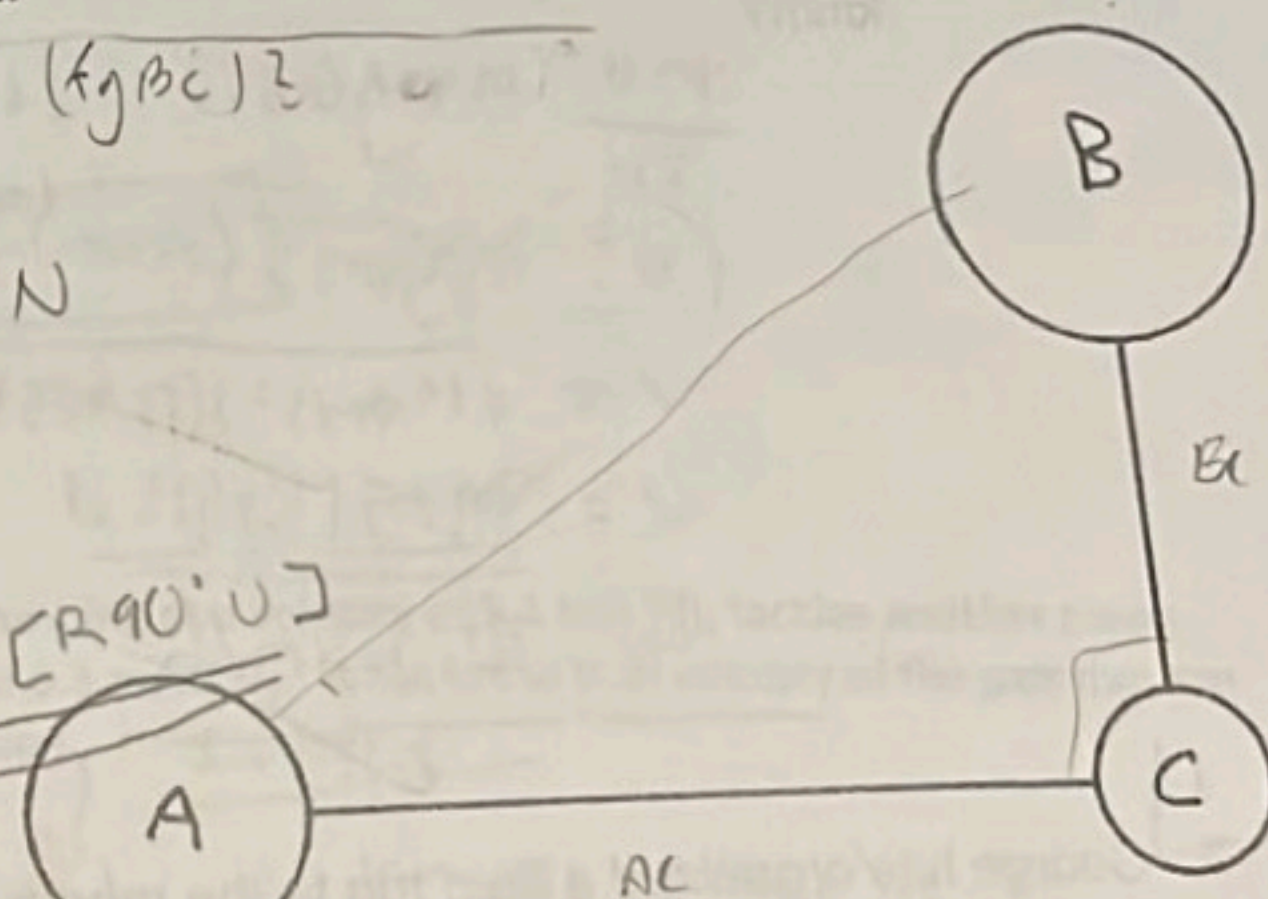
$$= \frac{(6.673 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.0 \times 10^{20} \text{ kg})(3.0 \times 10^{20} \text{ kg})}{(9.0 \times 10^{13} \text{ m})^2}$$

$$F_{gAC} = 1235.740 \text{ N [R]}$$

$$F_{gBC} = \frac{G m_B m_C}{r_{BC}^2}$$

$$= \frac{(6.673 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(1.2 \times 10^{21} \text{ kg})(3.0 \times 10^{20} \text{ kg})}{(6.5 \times 10^{13} \text{ m})^2}$$

$$F_{gBC} = 5.68 \times 10^{-7} \text{ N [U]}$$



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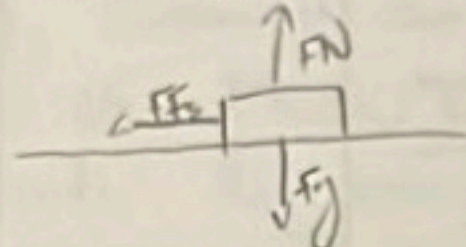


2. Julia is a civil engineer who is creating a new high-friction pavement for a roundabout in Richmond Hill. The speed limit in the area is 50 km/h, and the roundabout has a diameter of 42 m.

a. What should the coefficient of friction be on the new pavement?

$$V = 13.8 \text{ m/s}$$

$$r = 21 \text{ m}$$



$$F_g = 0$$

only motion on x

$$F_c = F_x$$

$$F_c = F_N \mu_s$$

$$\frac{mv^2}{r} = mg \mu_s$$

$$\frac{v^2}{rg} = \mu_s$$

$$\frac{(13.8 \text{ m/s})^2}{(21 \text{ m})(9.8 \text{ m/s}^2)} = \mu_s$$

$$0.925 = \mu_s$$

- b. If the town doubled the available space for the roundabout (r) but the type of pavement stayed the same, what could the new theoretical safe speed limit be in km/h?

$$\frac{mv^2}{2r} = mg \mu_s$$

$$\sqrt{v^2} = \sqrt{g \mu_s 2r}$$

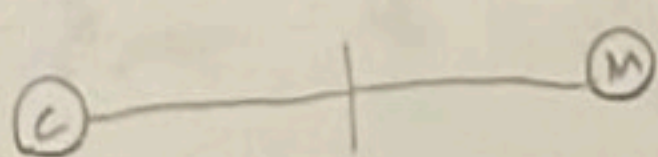
$$v = \sqrt{(9.8 \text{ m/s}^2)(0.925) 2(21 \text{ m})}$$

$$v = 19.51 \text{ m/s}$$

$$\rightarrow 70.2 \text{ km/h}$$

3. George has organized a field trip to the moon. When the class gets part way there, they run out of fuel and end up in orbit around the moon ($7.35 \times 10^{22} \text{ kg}$) at 360 km/h.

- a. How far away from the centre of the moon did George and the class get?
b. What is the frequency and period of the orbit?



$$V = 100 \text{ m/s} \quad m = 7.35 \times 10^{22} \text{ kg}$$

$$b) T = \frac{2\pi r}{v}$$

$$= \frac{2\pi (4.90 \times 10^9 \text{ m})}{(100 \text{ m/s})}$$

$$T = 30816856.23 \text{ s}$$

$$T = 30800000 \text{ s}$$

$$f = \frac{1}{T}$$

$$= \frac{1}{30816856.23 \text{ s}}$$

$$f = 3.24 \times 10^{-6} \text{ Hz}$$

$$V = \sqrt{\frac{GM}{r}}$$

$$V^2 = \frac{GM}{r}$$

$$r = \frac{GM}{V^2}$$

$$= \frac{(6.673 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2)(7.35 \times 10^{22} \text{ kg})}{(100 \text{ m/s})^2}$$

$$r = 4.90 \times 10^9 \text{ m}$$



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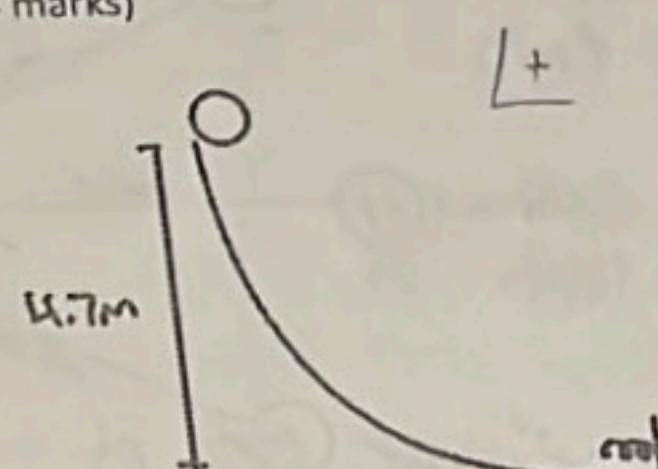
Chapter 4 and 5 Energy and Momentum Unit Test

Name: [REDACTED]

Thinking/Inquiry

Clearly show ALL of your work for full credit. All questions on this test must be answered using energy and momentum concepts where possible. This includes the X and Y component Method!!! Answers that do not use energy and momentum concepts will not receive full marks. Round your final answer to 3 sd.

1. A ball (2.1 kg) at the top of the ramp is traveling straight down with a velocity of 1.8 m/s. The ramp is 4.7 m tall. How far will the spring compress if it has a k value of 35 N/m? (4 marks)



$$E_{mi} = E_{mf}$$

$$E_{gi} + E_{ki} + E_{ei} = E_{gf} + E_{kf} + E_{ef}$$

$$mgh_i + \frac{1}{2}mv_i^2 = \frac{1}{2}kx^2$$

$$\sqrt{\frac{mgh_i + \frac{1}{2}mv_i^2}{2k}} = x$$

$$\sqrt{\frac{(2.1\text{kg})(9.8\text{m/s}^2)(4.7\text{m}) + \frac{1}{2}(2.1\text{kg})(1.8\text{m/s})^2}{2(35\text{N/m})}} = x$$

$$1.20\text{m} = x$$

2. During a Rugby game, George of mass 84 kg, running at a velocity of 9.1 m/s [E], tackles another player head-on of mass 95 kg, running at a velocity of 6.3 m/s [W]. What is the final velocity of the pair that are joined together just after the collision? (3 marks) Inelastic

$$m_1 = 84\text{kg}$$

$$v_{1i} = 9.1\text{m/s [E]}$$

$$m_2 = 95\text{kg}$$

$$v_{2i} = 6.3\text{m/s [W]}$$

$$v_f = ?$$

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_f + m_2 v_f$$

$$\frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2} = \frac{v_f (m_1 + m_2)}{m_1 + m_2}$$

$$v_f = \frac{(84\text{kg})(9.1\text{m/s}) + (95\text{kg})(-6.3)}{179\text{kg}}$$

$$v_f = 0.927\text{m/s [E]}$$



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3. Julia and Chloe are dissecting a brain (1.4 kg) of her pig Bartholomew in biology class when Julia gets grossed out by the cerebellum. She accidentally throws Bartholomew's brain east towards Chloe at a speed of 8.0 m/s, but Chloe reacts and saves it from hitting the ground by kicking it right back to the dissection tray at a speed of 10 m/s. If Chloe's foot made contact for 1.7×10^{-1} s, what was the average force of the kick? (3 marks)

$$v_{1i} = 8 \text{ m/s}$$

$$v_{1f} = 8 \text{ m/s}$$

$$v_{2i} = 8 \text{ m/s}$$

$$v_{2f} = 10 \text{ m/s}$$

$$Fb = mv_f - mv_i$$

$$F = m(v_f - v_i)$$

$$F = (1.4 \text{ kg}) \frac{(-10 \text{ m/s} - (8 \text{ m/s}))}{1.7 \times 10^{-1} \text{ s}}$$

$$F = -148 \text{ N (W)}$$

∴ the average force of the kick was 148 N (W)

4. Block 1 (2.2 kg) is at the top of a ramp angled at 30° above the horizontal. Block 1 collides perfectly elastically with block 2 (7.0 kg). If block 2 slides up the second ramp to a height of 3m, how far does the spring compress? (5 marks)

$$m_1 = 2.2 \text{ kg}$$

$$\theta = 30^\circ$$

$$m_2 = 7 \text{ kg}$$

$$\textcircled{1} E_m = E_g$$

$$= mgh$$

$$= (2.2 \text{ kg})(9.8 \text{ m/s}^2)(15 \text{ m})$$

$$E_m = 323.4 \text{ J}$$

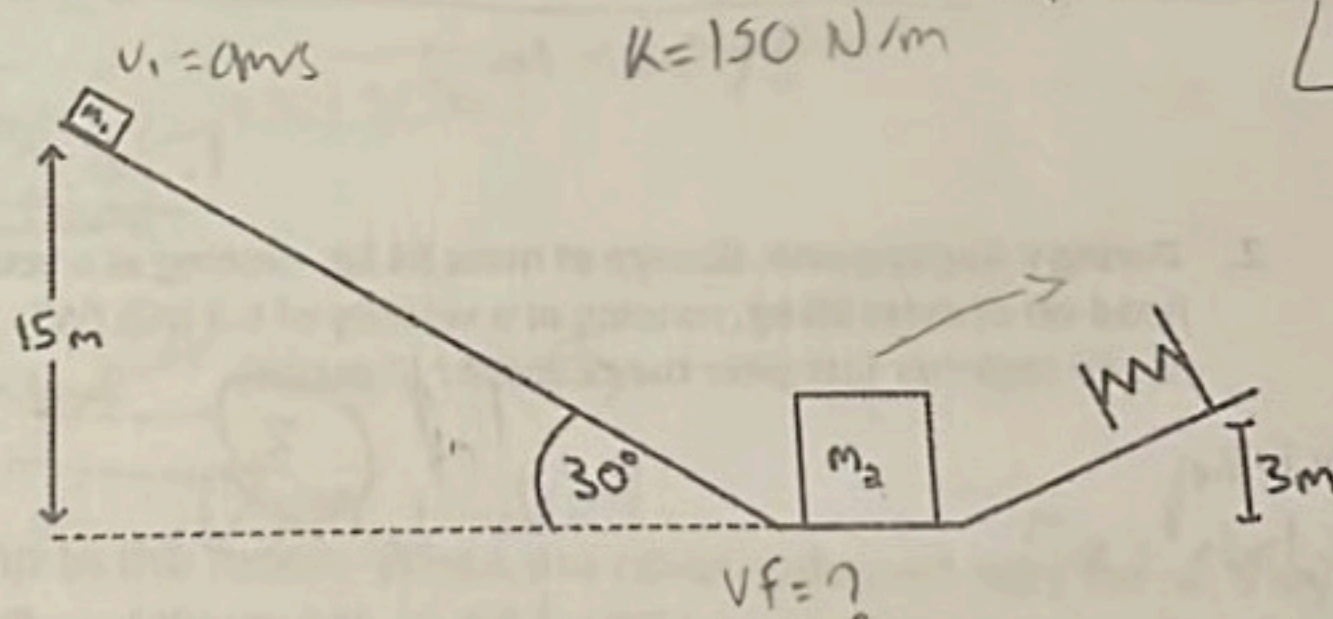
$$\textcircled{2} E_m = \Delta E_k$$

$$E_m = \frac{1}{2} m_2 v_f^2$$

$$\sqrt{\frac{2 E_m}{m}} = v_f$$

$$\sqrt{\frac{2(323.4 \text{ J})}{7 \text{ kg}}} = v_f$$

$$9.61 \text{ m/s} = v_f$$



$$\textcircled{3} E_{m1} = E_{m2}$$

$$E_{g1} + E_{k1} + E_{e1} = E_{g2} + E_{k2} + E_{e2}$$

$$\frac{1}{2} m_1 v_i^2 = m_2 g h_2 + \frac{1}{2} k x^2$$

$$\sqrt{\frac{\frac{1}{2} m_1 v_i^2 - m_2 g h_2}{\frac{1}{2} k}} = x$$

$$x = \sqrt{\frac{\frac{1}{2} (7 \text{ kg}) (9.61 \text{ m/s})^2 - (7 \text{ kg}) (9.8 \text{ m/s}^2) (3 \text{ m})}{\frac{1}{2} (150 \text{ N/m})}}$$

$$x = 0.626 \text{ m}$$



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